

Emergency Review and Constitutional Court Design by Simulation

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Abstract

Constitutional courts face acute legitimacy pressure when emergency orders resolve public-law disputes before ordinary merits review. This paper uses a shared-world simulation to compare emergency-review rules inside broader constitutional-court design bundles. Across twenty-seven designs and twenty assumption cases, written reasons, automatic merits follow-through, and no-relief-without-merits review mechanically regularize emergency procedure and, under the reported assumptions, also reduce downstream emergency disruption. Those gains are not costless: domain-specific claimant success, lower-court compliance, implementation risk, and administrative cost remain central to interpretation. Appointment, panel, council, override, and judicial-electorate reforms mainly shift legitimacy, access, compliance, administrative-cost, or conflict channels rather than directly regularizing emergency review. The results are simulation diagnostics under stated assumptions, not estimates of real-world causal effects.

Keywords: constitutional review, judicial politics, institutional design, simulation, courts

1 Design Question

The central question is how emergency-review rules alter constitutional-court tradeoffs when they operate inside a broader institutional bundle. A court can be independent while losing democratic legitimacy, responsive while failing rights protection, stable while entrenching poor doctrine, or active while creating constitutional conflict. Emergency orders intensify those tensions because they can preserve rights or government authority before ordinary merits review, often with less time, less reason giving, and less visible procedural regularity.

This makes the paper a computational institutional-design exercise for judicial politics and comparative public law. Emergency procedure is the main object of analysis: written reasons, merits follow-through, full-court referral, expiration, and no-relief-without-merits rules. Appointment regularity, admission filtering, recusal, panel routing, override rules, council warnings, cross-court checks, and post-decision compliance strategies remain in the model because they shape the costs and credibility of emergency relief.

The U.S. Supreme Court supplies the principal calibration anchor for certiorari and emergency-procedure guardrails, but it is not the model's jurisdictional limit. Comparative constitutional-review devices enter as design mechanisms to test whether emergency-process reforms remain distinguishable from appointment, access, council, override, panel, or compliance reforms.

2 Contribution

The paper makes three contributions. First, it treats emergency-review reform as a bundle-level design problem rather than as a transparency proposal in isolation. A written-reasons rule, for example, interacts with admission pressure, merits acceleration, recusal, lower-court compliance, and post-decision government response. Second, it reports direct procedural and outcome outputs before

constructed indices, so emergency grants, claimant success, domain-specific claimant outcomes, lower-court compliance, government noncompliance, emergency downstream effects, and process legitimacy remain visible. Third, it provides a reproducible design-search workflow that links scenario definitions, normalized plausibility inputs, deterministic seeds, and manuscript outputs.

These contributions are methodological rather than doctrinal. We do not argue that simulation can settle constitutional design. We argue that explicit simulation can make institutional tradeoffs more inspectable, especially when reforms that look attractive in isolation shift costs into emergency procedure, admission filtering, public legitimacy, interbranch conflict, or post-decision compliance.

3 Claim Boundaries and Interpretation

The manuscript separates empirical claims, synthetic findings, and speculative design recommendations. Empirical claims are limited to source descriptions, normalized plausibility checks, cited literature, and reproducible project artifacts. They support statements such as which source families are used for plausibility checks, what institutional features are represented in the code, and whether generated artifacts match their manifests. They do not establish causal effects of any reform in the world.

Synthetic findings are outputs of the simulator under stated coding choices, deterministic seeds, and assumption cases. Statements about a scenario reducing emergency irregularity, increasing administrative cost, changing lower-court compliance, or moving the process-legitimacy index are therefore simulation findings, not empirical findings about observed courts. The generated tables and figures show where costs move when a rule bundle changes.

Speculative design recommendations are conditional inferences from those synthetic findings. The paper does not recommend weakening or strengthening courts as an end in itself. The design objective used here is narrower: reduce arbitrary emergency power while preserving rights protection, visible legal reasoning, lower-court settlement, and credible implementation. Reforms are therefore evaluated against noncompliance, recusal pressure, and rights-failure tradeoffs as well as emergency restraint.

4 Theory and Design Space

The model joins three law-and-courts traditions that are often studied separately. The first treats judges as purposive actors whose choices depend on legal goals, policy preferences, colleagues, litigants, and other institutions (Segal and Spaeth, 2002; Epstein and Knight, 1998; Clayton and Gillman, 1999). The second treats constitutional review as an institutional settlement among political actors who face uncertainty, seek insurance against future exclusion, or try to preserve influence through courts (Ginsburg, 2003; Hirschl, 2004; Dixon and Ginsburg, 2017). The third asks how courts preserve authority when they depend on elected officials, lower courts, publics, and repeat players for implementation (Vanberg, 2005; Staton, 2010). A useful simulator for constitutional-review design has to represent all three: judicial choice, institutional allocation, and post-decision compliance.

Comparative constitutional design also makes clear that “judicial review” is not one institution. Strong-form review, weak-form review, pre-enactment review, legislative override, individual complaint, amparo, concrete referral, constitutional council screening, specialized constitutional courts, and ordinary supreme courts expose different actors to different veto points (Tushnet, 2008; Garbaum, 2013; Sweet, 2000; Brewer-Carías, 2009; Kommers and Miller, 2012). Override and weak-form review scholarship also shows that post-decision political response is part of the institution

rather than an afterthought (Hogg and Bushell, 1997; Comella, 2009). That is why this project models access path, review timing, remedy threshold, emergency procedure, and override rule as separate design fields. The relevant unit of comparison is the bundle, not a single reform label.

Emergency procedure is treated separately because it changes both timing and legitimacy. The literature on the U.S. shadow docket emphasizes that orders outside the ordinary merits process can be consequential precisely because they are faster, less reasoned, and less procedurally regular (Baude, 2015; Vladeck, 2023; Kastellec and Taboni, 2026). The simulator therefore separates ordinary merits review from emergency applications, emergency relief, public disagreement, reasons, merits follow-through, and status-quo effects. This makes emergency legality, rights protection, and legitimacy costs visible instead of burying them in an aggregate activity rate.

The design problem also differs from ordinary policy optimization. Constitutional-review institutions produce several goods that can conflict: rights protection, legal stability, responsiveness to democratic majorities, settlement of interbranch conflict, and visible decisional legitimacy. A design that improves one good can undermine another by changing access, delay, information, coalition structure, or implementation incentives. For that reason, the simulator does not search for a single welfare-maximizing court. It asks whether a design bundle moves costs into predictable locations and whether those costs remain visible.

This framing matters for reform debates. A proposal to impose term limits is not only a tenure proposal; it changes vacancy timing, appointment incentives, and the value of partisan capture. A proposal to require reasoned emergency orders is not only a transparency proposal; it changes speed, emergency grant behavior, public disagreement, and merits follow-through. A proposal to add a constitutional council or judicial council is not only a rights-screening proposal; it changes *ex ante* review, appointment insulation, later judicial workload, and the political salience of invalidation (Garoupa and Ginsburg, 2009). The simulation architecture therefore treats legal form, access path, and political response as jointly produced institutional behavior.

5 Expectations

The simulation study is organized around five design expectations. Each links an institutional rule to the outputs most likely to reveal its tradeoffs.

First, emergency-reason-giving and merits-follow-through rules should reduce emergency irregularity and emergency legitimacy risk. The relevant tradeoff metrics are emergency downstream effects, lower-court compliance, rights-claimant success, and administrative cost.

Second, no-relief-without-merits and expiration rules should reduce merits-displacing emergency intervention. The tradeoff test is whether they preserve rights protection when emergency relief would otherwise be the claimant’s only timely remedy.

Third, appointment and selection reforms should mostly affect partisan alignment, appointment-pressure campaigns, and process legitimacy rather than emergency-process outputs. Term limits, commissions, and judicial-electorate selection provide comparison cases for distinguishing emergency procedure from selection design.

Fourth, panel routing, cross-checking courts, dual supreme courts, and constitutional councils should not be treated as simple activity reducers. They may improve error correction, *ex ante* screening, or cross-institutional moderation, but they can also raise administrative cost, disagreement, or constitutional-conflict-index values if the rules for resolving institutional disagreement are weak.

Fifth, legislative override rules and accountability devices should increase democratic responsiveness only when they do not collapse into repeated override loops, rights carveout evasion, or symbolic compliance. The relevant output is therefore not only whether override is available, but

whether it changes practical implementation, rights protection, elite acceptance, and constitutional conflict.

6 Model

Each generated world contains a pool of potential justices, a filed universe of constitutional disputes, and a legislative-output profile. Filed cases vary along six grouped dimensions: legal strength, claimant capacity, political salience, emergency pressure, lower-court conflict, and implementation risk. Legal-domain profiles then shift those dimensions for rights, election, executive-power, administrative, structural, and economic-regulation disputes so the model does not treat all constitutional cases as interchangeable.

A review-path layer distinguishes certiorari-like petitions, direct constitutional complaints, amparo, QPC-style and concrete referrals, abstract review, emergency applications, electoral review, and interbranch disputes. Admission filtering turns filed disputes into screened-out matters or merits transfers using path-specific signals such as lower-court conflict, government participation, amici, relists, claimant resources, vehicle quality, repeat-player advantages, strategic plaintiff selection, settlement pressure, and conditional reversal probability. There is no preliminary fixed-size top-case selector before admission; the modeled merits docket is an output of the intake process. Figure 1 summarizes the state transition from filed matter to post-decision response. Appendix B describes the main mechanics; Table S4 in the technical supplement lists representative weights.

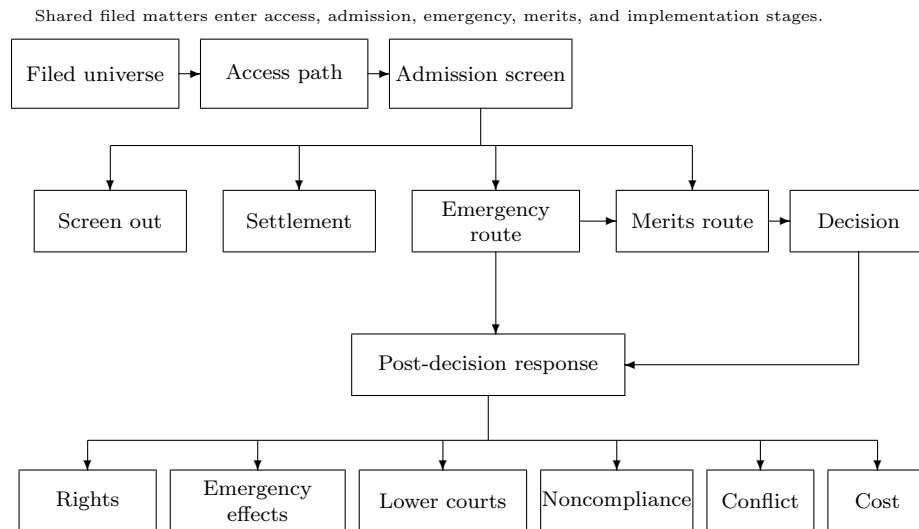


Figure 1: Core state transition for the shared-world simulator. The same filed dispute can be screened out, settled, routed to emergency disposition, transferred to merits review, or carried into post-decision implementation depending on access path and institutional design.

Each scenario supplies a court design. The design selects reviewers, applies recusal rules, routes cases to full-court or panel review, determines emergency relief procedure, applies voting thresholds, optionally invokes cross-checking or council review, and optionally allows legislative or popular override. Appointment rules include president-senate selection, nonpartisan commissions, legislative supermajority selection, lottery or rotating panels, and judicial-electorate selection. The

judicial-electorate mechanism separates the voting pool from the nominee pool, so the model can distinguish all federal judges, federal appellate judges, selected circuits, state high-court judges, and combined federal/state high-court electorates from narrower or broader nominee eligibility rules. Emergency applications carry applicant type, application class, requested relief, response request, full-court referral, status-quo effect, public disagreement risk, and merits follow-through. Emergency orders can also create downstream effects through status-quo disruption, opportunistic emergency filing, unexplained relief, merits follow-through, and later public disagreement. Strategic actors choose whether to comply, evade, reenact, flood emergency applications, organize override campaigns, or pressure appointments. Their response is split into formal legal response and practical implementation response, including repeal, replacement, narrowed reenactment, weak-form override, amendment, court-curbing, delay, administrative substitution, symbolic compliance, bureaucratic resistance, and open noncompliance.

The model uses shared worlds to make institutional comparisons sharper. Within a run, every scenario sees the same filed matters and legislative-output profile. Differences in outputs therefore come from the institutional rule bundle and its induced strategic responses, not from a different case draw. Across runs and stress cases, the world generator changes appointment capture, emergency pressure, rights risk, democratic mandate, public attention, executive defiance, and imported legislative quality. This gives the study both within-world comparison and across-assumption sensitivity.

The current design surface is deliberately broader than the tabled scenarios. Court size may be odd or even; terms may be life, fixed, staggered, renewable, or retirement-age bounded; removal may be impeachment-like, supermajoritarian, or external-review based; recusal may leave replacement, vacancy, or quorum consequences; remedies may require different thresholds for invalidation, emergency relief, or override; and judicial-electorate selection may vary both the voting electorate and the nominee eligibility pool. The scenario catalog includes randomized merits panels with en banc correction, mandatory written emergency reasoning, automatic merits follow-up, strong recusal enforcement, jurisdiction-stripping constraints, legislative override windows, constitutional remand, public-interest litigation filtering, a broad judicial-electorate selection design, and bundled systems that combine emergency integrity, remand-and-override, panel-and-jurisdiction, or council-and-concrete-review safeguards. The diagnostics vary assumptions around these bundles rather than presenting every possible combination as if it were substantively meaningful.

The twenty-seven reported designs fall into six families. Emergency-process designs change reason giving, merits follow-through, expiration, and relief-without-merits rules. Appointment and selection designs change term regularity, commission screening, legislative selection, court size, or judicial-electorate voting. Access and filtering designs change petition, public-interest, jurisdiction-stripping, or constitutional-complaint routes. Routing and correction designs change panels, en banc review, cross-checking courts, or constitutional councils. Post-decision designs change remand, override, rights carveouts, or implementation response. Bundled designs combine several of those mechanisms to test whether apparent emergency gains survive cost, compliance, and conflict tradeoffs.

That choice keeps the paper focused on institutional mechanisms rather than arbitrary parameter permutations.

An individual simulated dispute moves through the model as follows. First, the filed universe draws a legal domain and case type, then applies the imported legislative-output profile to the case's quality, mandate, rights burden, and conflict potential. Second, the access-path module assigns the procedural route. Third, the admission filter applies path-specific signals and either screens the matter out, settles it, or transfers it to emergency or merits review. Fourth, the scenario's court design determines reviewers, recusals, quorum effects, panel or en banc routing, voting thresholds,

reason giving, cross-court or council checks, and emergency merits follow-through. Fifth, the decision creates a formal legal response, a practical implementation response, a precedent-durability value, and an emergency downstream effect. Enforcement capacity, lower-court resistance, lower-court compliance, and elite acceptance then feed the process-legitimacy and constitutional-conflict indices.

7 Calibration and Plausibility Checks

Calibration is limited to plausibility checks. Normalized source observations are drawn from the Supreme Court Database, Harvard Law Review Supreme Court statistics, the Supreme Court Shadow Docket Database, Black and Epstein’s recusal data, and official Supreme Court Journal and docket records (Supreme Court Database, 2025; Harvard Law Review, 2025; Kastellec and Taboni, 2026; Black and Epstein, 2005; Supreme Court of the United States, 2024, 2025, 2026a,b). Comparative synthesis rows generated during project research serve as structured design-context inputs with source names, URLs, confidence labels, validation-use fields, denominator notes, and coverage scopes. Source rows are classified as strict validation, loose calibration, proxy, mechanism, or design-prior checks. Only strict-validation rows approximate source denominators; proxy and mechanism rows are used as plausibility checks.

For certiorari-stage screening, the replication data independently enumerate Journal-count-defined paid and in-forma-pauperis docket ranges for OT2023 and OT2024. The two cohorts contain 8,076 dockets and 7,716 certiorari petitions. Three OT2024 petitions remained pending or held at the July 26, 2026 snapshot, so that term’s grant/GVR rate uses 3,680 resolved petitions. Table 1 preserves each term’s numerator and denominator. The largest two-term movements are a 2.93 percentage-point decline in relist presence and a 2.04-point increase in Court-requested responses among paid petitions. With only two terms, those changes describe short-window variation rather than a trend, stable population parameter, or causal effect.

A separate field audit distinguishes populated entries from substantive coding. The 30 CVSG-requested certiorari petitions have filing markers but no coded recommendation direction; an additional CVSG row belongs to non-certiorari intake. Unresolved court-origin and actor categories likewise remain incomplete. The replication materials preserve a full petition coding queue and a fixed 40-petition pilot selected across term and paid/IFP strata without using outcomes. AI-assisted extraction and audit scripts produce deterministic artifacts from frozen source inputs; automated legal-treatment candidates remain subject to independent human review. These are acquisition priorities, not new behavioral observations.

Table 1: Official-docket certiorari-stage benchmarks across OT2023 and OT2024

Measure	OT2023	OT2024	Δ pp	Pooled
Paid share of paid/IFP intake	32.57%	34.43%	+1.86	33.46%
Court-requested response, paid	12.58%	14.63%	+2.04	13.59%
Court-requested response, IFP	2.57%	3.14%	+0.57	2.83%
CVSG request	0.25%	0.54%	+0.30	0.39%
Cert-stage amicus presence	7.56%	8.50%	+0.94	8.01%
Relist presence	14.03%	11.11%	-2.93	12.64%
Grant or GVR among resolved certiorari petitions	3.35%	3.10%	-0.25	3.23%

Each term is an independently enumerated paid/IFP published-statistics snapshot cohort. Pooled rates sum term numerators and denominators. The comparison is descriptive, excludes undocketed submissions, and is not limited to constitutional-review petitions. The grant/GVR row excludes three OT2024 petitions still pending or held as of the snapshot. The OT2024 manifest separately discloses same-cutoff-date dockets above the Journal count-defined ranges.

These observations cover docketed intake only. They exclude undocketed submissions, are not limited to constitutional-review petitions, and do not establish issue mix, specialist-counsel, or split-quality patterns. The OT2024 cohort is a published-statistics snapshot: its manifest separately discloses four same-cutoff-date public dockets above the Journal count-defined ranges.

A separate direct aggregate benchmark measures doctrinal uptake rather than docket intake. The public replication file for Masood, Kassow, and Songer contains 876 formally argued Supreme Court precedents from the 1995–2004 terms, including 861 rows eligible for the article’s main models, and accumulates lower-court Shepard’s treatment counts through 2016 (Masood et al., 2019a,b). Across all 876 precedents, the source records 235,256 followed treatments and 38,297 adverse or distinguished treatments, yielding an 86.0% followed share among directional treatments. Among the 223 source-flagged constitutional-issue precedents, 47,370 followed and 17,753 adverse treatments yield a 72.7% directional followed share.

Those shares are bounded doctrinal-uptake descriptions, not practical-compliance rates. The released unit is a Supreme Court precedent, not an individual lower-court opinion; aggregate counts weight heavily treated precedents more heavily, omit exposed but non-citing or ignored cases, and do not disclose treatment dates, lower-court identifiers, remedies, or implementation events. The constitutional subset’s directional share is therefore shown as proxy context with a scale mismatch to the simulator’s synthetic case-average `lowerCourtCompliance` score, not as its validation target.

A separate environmental cohort supplies a bounded descriptive event layer. Five decision-specific CourtListener searches explicitly select published federal circuit and district opinions filed during the 730 days after *Rapanos*, *Massachusetts*, *Utility Air Regulatory Group*, *Michigan*, and *Sackett* (Free Law Project, 2026). After provider-cluster deduplication with a stable cluster-identifier tie-breaker, the extract contains 191 citation-linked opinion documents and a 65-cell decision-by-circuit frame for nationwide precedential applicability and published-citation presence. Public full text and source-decision context are available for 115 documents, but missingness varies sharply by decision and opinion-document type. Conservative target-linked rules flag five applied candidates and one distinguished candidate; 109 are citation-only and 76 remain unclear. The directional values remain pending expert legal review. A zero-citation cell records no observed published citation, not empirical exposure, ignored precedent, or noncompliance.

The practical layer preserves the complete purposive five-case sample and three-part agency-response classifications in Gurganus’s study of salient environmental decisions (Gurganus, 2025). Four episodes are classified as compliant and *Rapanos* as narrowly compliant, with each row joined to an official agency or Federal Register action and a structured Table 1 transcription. The sample is statutory, salient, purposive, and contains no noncompliant outcome. Its four-to-one composition describes selected case studies, not a compliance probability, general resistance rate, or target for `governmentNoncomplianceRate`.

Table 2 reports representative guardrails rather than every source row. It shows the type of check being applied: a current-like design should have plausible docket composition, invalidation, recusal, and emergency-order behavior, while a no-emergency-relief-without-merits design should sharply suppress emergency relief without erasing emergency merits acceleration.

Table 3 is a safeguard against overstating the evidence. Strict-validation rows are closest to empirical denominators, loose-calibration rows are plausibility checks, proxy rows are warning lights, mechanism rows test whether coded mechanisms move in the expected direction, and model-prior rows document institutional coding assumptions. A scenario does not become empirically validated because it passes a proxy or mechanism row.

Table 4 adds a quality audit for representative rows. It reports range width and distance from source median so that a broad interval does not masquerade as strong validation. Rows with proxy, mechanism, or design-prior uses remain useful guardrails, but their passing status should not carry

Table 2: Selected calibration guardrails

Guardrail	Scenario	Use	Evidence	Observed	Range	Source basis	Status
Merits transfer	Current-like	model prior check	research synthesis	0.390	0.250–0.850	source register; model fallback	within
Paid CFR stage	Current-like	loose calibration	raw or primary summary	0.045	0.022–0.171	research tables + official dockets; source range	within
IFP CFR stage	Current-like	loose calibration	raw or primary summary	0.039	0.000–0.051	research tables + official dockets; source range	within
Invalidation	Current-like	strict validation	raw or primary summary	0.090	0.000–0.223	SCDB; source range	within
Emergency filings	Current-like	proxy sanity check	raw or primary summary	0.046	0.000–0.223	shadow database; source range	within
Emerg. irregularity	Current-like	loose calibration	raw or primary summary	0.209	0.000–0.657	shadow database; source range	within
Partisan alignment	Commission	proxy sanity check	raw or primary summary	0.020	0.000–0.657	shadow database; source range	within
Emergency restraint	No-gap	mechanism sanity check	raw or primary summary	0.007	0.000–0.657	shadow database; source range	within

Table 3: Calibration-use classification summary

Guardrail use	Rows	Within range	Interpretation
loose calibration	11	11	plausibility range, not point validation
mechanism sanity check	2	2	context row
model prior check	2	2	context row
proxy sanity check	4	4	context row
strict validation	4	4	closest to source-denominator checks

the evidentiary weight of a denominator-matched validation target.

Table 5 is the denominator audit. Certiorari screening, emergency applications, individual complaints, QPC/concrete referrals, lower-court compliance, and override/remand behavior are shown as separate source-comparison pathways. The table records both the manuscript-use label and the denominator compatibility judgment because a source row that is strict within its own dataset can still be only proxy context for a simulator metric.

8 Metrics

The paper-facing metrics are grouped into four categories. Direct procedural outputs include petition filing, court-requested response, CVSG request, admission, emergency order, emergency grant, reasoned emergency order, merits acceleration, recusal, panel, en banc, council warning, remand, public-interest filtering, and override rates. Direct outcome outputs include claimant success, invalidation, precedent durability, lower-court compliance, government noncompliance, elite acceptance, emergency downstream effect, and domain-specific claimant success. Constructed indices include emergency irregularity, emergency legitimacy risk, the constitutional-conflict index, and the process-legitimacy index. Normative reading aids include the directional score and alternative

Table 4: Calibration-quality audit for selected plausibility checks

Check	Sim. value	Source median	Full range	Range width	Distance	Denominator quality	Use	Failure rule
Rights-related merits domains track SCDB rights issue areas.	0.204	0.299	0.064–0.479	0.415	0.095	near source denominator	strict validation	outside source range
Administrative-law challenges track SCDB administrative-action observations.	0.184	0.234	0.000–0.503	0.503	0.050	near source denominator	strict validation	outside source range
Admissibility-aware current-like designs should transfer a substantial but not universal share of filed matters to merits.	0.390	0.550	0.250–0.850	0.600	0.160	design prior	model prior check	coding review
Declarations of unconstitutionality should be uncommon in the full merits docket.	0.090	0.066	0.000–0.223	0.223	0.024	near source denominator	strict validation	outside source range
Emergency applications should be present but bounded in the current-like docket; the source denominator is Journal orders, not generated filed matters.	0.046	0.006	0.000–0.223	0.223	0.040	proxy only	proxy sanity check	proxy warning
Open emergency procedure creates measurable but bounded shadow-docket abuse.	0.209	0.222	0.000–0.657	0.657	0.013	broad source range	loose calibration	outside broad range
Justice-case recusals should be rare.	0.013	0.009	0.000–0.029	0.029	0.004	near source denominator	strict validation	outside source range
No-relief-without-merits should sharply suppress shadow-docket abuse.	0.007	0.222	0.000–0.657	0.657	0.215	mechanism proxy	mechanism sanity check	wrong direction

Wide ranges and proxy rows are reported as calibration weaknesses, not as stronger evidence merely because the simulated value falls inside the interval.

Table 5: Pathway-specific denominator audit

Pathway	Construct	Sim. metric	Current	Source metric	Use	Compatibility	Next action
certiorari	paid certiorari intake share	paid cert petition share	0.325	paid petition share	loose calibration	near pathway match	near match
certiorari	paid court-requested response	paid cfr request rate	0.061	cfr rate paid	loose calibration	near pathway match	near match
certiorari	grant/admission funnel	certiorari admission rate	0.204	us scotus plenary review rate all docketed	proxy context	denominator mismatch	validation target needed
emergency	emergency application presence	emergency stay docket rate	0.092	emergency stay docket rate	proxy context	denominator mismatch	validation target needed
emergency	emergency relief grants	emergency grant conditional rate	0.241	noncapital grant rate overall	loose calibration	conditional near match	near match
complaint referral	individual complaint admission	admission rate	0.538	spain amparo admission rate	proxy context	denominator mismatch	validation target needed
lower court compliance	lower-court doctrinal uptake	lower court compliance	0.513	lower court followed share among directional treatments constitutional issues	proxy context	scale mismatch	proxy only
lower court compliance	government noncompliance	government noncompliance rate	0.183	federal agency narrow compliance share	loose calibration	mechanism proxy	mechanism proxy
complaint referral	constitutional remand/deferred remedy	constitutional remand rate	0.000	fr qpc delayed effect rate decided qpc	proxy context	mechanism proxy	mechanism proxy
override remand	override pressure after invalidation	override attempt rate	0.000	invalidation rate	proxy context	mechanism proxy	mechanism proxy

priority scores in Table S1 in the technical supplement.

Emergency irregularity measures opaque, opportunistic, or merits-displacing emergency intervention. Process legitimacy is a modeled index based on visible procedure, emergency risk, partisan alignment, appointment pressure, and court-curbing response. Rights-claimant success is reported separately from rights protection and disaggregated by legal domain.

The directional score is an appendix reading aid. It averages a stability/rights score, a legitimacy/control score, rights-claimant success, precedent durability, lower-court compliance, elite

acceptance, and administrative feasibility. Because that is a constitutional theory rather than a neutral measurement choice, the main results do not order designs by the directional score. Appendix Table 17 records the denominator or scale for representative metrics.

When a reform directly changes a component used in an index, movement in that index is treated as a design implication rather than a standalone discovery. The analysis emphasizes cases where downstream measures also move, including rights-claimant success, emergency downstream effects, lower-court compliance, government noncompliance, constitutional conflict, or administrative cost.

8.1 Non-Mechanical Diagnostics

The central diagnostic question is whether emergency-process rules move downstream disruption, rights-claimant success, lower-court compliance, government noncompliance, and administrative cost after the direct procedural regularization is set aside. Table 6 reports that check for the main emergency-review designs. Emergency irregularity remains a diagnostic of procedural regularization; the downstream-only column deliberately excludes emergency irregularity and process legitimacy.

Table 6: Non-mechanical downstream diagnostics for emergency-review reforms

Scenario	Reported irregularity	No reasons credit	No accel. credit	Downstream	Rights claimants	Lower ct.	Gov. noncomp.	Downstream-only
Current-like	0.308	0.308	0.308	0.187	0.191	0.513	0.183	0.636
Written reasons	0.077	0.110	0.087	0.129	0.157	0.535	0.149	0.637
Auto merits	0.030	0.063	0.059	0.089	0.212	0.541	0.150	0.654
No merits gap	0.022	0.055	0.051	0.088	0.207	0.543	0.148	0.657
Integrity package	0.030	0.064	0.059	0.090	0.209	0.543	0.148	0.649

The two ablation columns recalculate the irregularity index with the direct reason-giving or merits-acceleration credit removed from the reported aggregate; the downstream-only score excludes emergency irregularity and process legitimacy.

Process legitimacy is treated with similar caution because it includes visible-procedure, emergency-risk, emergency-irregularity, partisan-alignment, appointment-pressure, and court-curbing channels. Table 7 recalculates the index after removing selected penalty families and adds an implementation-only comparison. The point is to show how much of the legitimacy pattern depends on directly coded emergency and partisan channels rather than on downstream implementation behavior.

Table 7: Process-legitimacy penalty ablations

Scenario	Reported	No irregularity penalty	Half irregularity penalty	No emergency-risk penalty	No partisan penalty	Implementation-only
Current-like	0.345	0.419	0.382	0.402	0.362	0.623
Written reasons	0.490	0.509	0.499	0.534	0.507	0.630
Auto merits	0.547	0.554	0.550	0.586	0.563	0.628
No merits gap	0.573	0.578	0.576	0.613	0.585	0.632
Recusal enforce.	0.520	0.546	0.533	0.563	0.531	0.629
Integrity package	0.577	0.585	0.581	0.618	0.588	0.624

Recalculations are aggregate penalty ablations, not new case-level reruns. They test whether the visual process-legitimacy pattern depends mainly on the directly coded emergency and partisan penalty channels.

Figure 2 illustrates why the domain-specific claimant-success measures are kept separate. Designs can look similar on aggregate rights protection while moving success rates differently for rights, structural, election, executive-power, administrative-law, and economic-regulation disputes.

9 Emergency-Review Results

The simulation study compares twenty-seven institutional scenarios across twenty assumption cases: the original broad cases, sensitivity sweeps for appointment capture, emergency pressure, rights risk, and democratic mandate, plus adversarial stress cases for appointment timing, emergency-application

	Rights	Struct.	Elect.	Exec.	Admin.	Econ.
Current-like	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00
18-year terms	-0.02	-0.02	-0.23	-0.13	-0.08	-0.01
Written reasons	-0.06	-0.03	-0.23	-0.13	-0.12	-0.03
Auto merits	+0.06	+0.07	+0.00	+0.02	+0.07	+0.04
No merits gap	+0.06	+0.03	-0.02	-0.03	+0.04	+0.02
Recusal enforce.	-0.02	-0.03	-0.23	-0.11	-0.07	-0.02
Random panels	-0.02	-0.02	-0.22	-0.13	-0.09	-0.01
Integrity pkg.	+0.06	+0.04	-0.01	-0.01	+0.04	+0.02

Change from current-like claimant success; black is higher, red is lower.

Figure 2: Domain-specific claimant-success changes relative to the current-like baseline. The aggregate rights-protection score can hide domain shifts, especially for election, structural, executive-power, administrative-law, and economic-regulation cases.

flooding, override evasion, recusal pressure, and court-expansion retaliation. The reported run used 80 runs per case, 64 filed matters per run, seed 20260501, and an imported legislative-output profile. Because written reasons, merits follow-through, and merits acceleration are built into the emergency-procedure rules, the main result is not that those rules mechanically lower the emergency-irregularity diagnostic. The more important result is that several emergency-process designs also reduce downstream emergency disruption while preserving aggregate rights outputs under the baseline assumptions, though with administrative-cost and domain-specific claimant-success tradeoffs. Appointment, panel, council, override, and judicial-electorate reforms mainly shift legitimacy, access, compliance, administrative-cost, or conflict channels.

Figure 3 is the main visual result. Similar rights and stability averages can coexist with very different emergency profiles, so the manuscript reports emergency irregularity, emergency legitimacy risk, and process legitimacy separately.

Table 8: Emergency-application-flood walkthrough under a shared assumption case

Design	Emerg. orders	Reasoned orders	Merits accel.	Irregularity	Downstream	Rights claimants	Emerg.-rights	Lower ct.	Cost	Process legit.
Current-like	0.647	0.000	0.000	0.514	0.388	0.275	0.978	0.403	0.191	0.239
Written reasons	0.636	0.636	0.133	0.169	0.290	0.144	0.517	0.436	0.286	0.432
Auto merits follow-up	0.644	0.644	0.644	0.072	0.206	0.277	0.988	0.445	0.307	0.560
No emergency merits gap	0.635	0.635	0.635	0.059	0.205	0.269	0.981	0.451	0.285	0.594

Table 8 gives the simplest shared-world intuition. Under the emergency-application-flood assumption case, written reasons, automatic merits follow-up, and no-relief-without-merits rules alter the route from emergency filing to reasons, merits acceleration, emergency irregularity, downstream disruption, rights-claimant outcomes, lower-court compliance, administrative cost, and process legitimacy. The emergency-rights column is included because a procedure that suppresses emergency relief would be unattractive if it preserved regularity only by denying urgent rights claimants a usable route.

Table 9 reports selected diagnostics with movement labels relative to the current-like baseline. The stylized current-like court has substantial modeled rights protection in this run while also

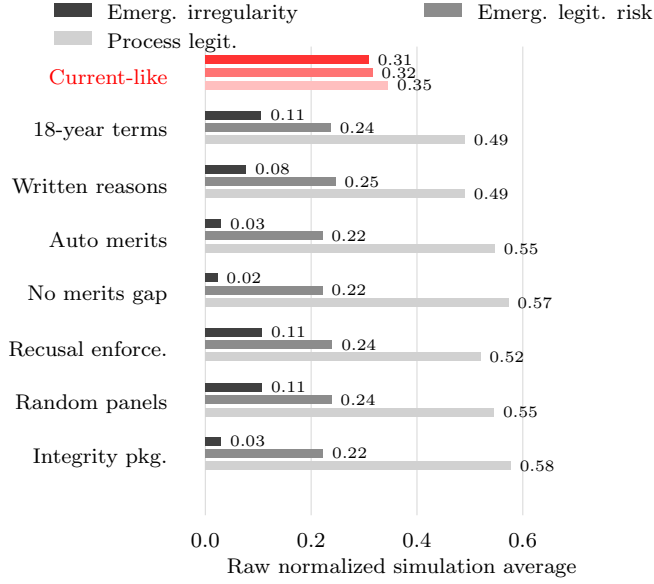


Figure 3: Emergency-review profile for selected institutional designs. Designs that are similar on rights protection and legal stability can differ sharply on emergency irregularity, emergency legitimacy risk, and process legitimacy. Emergency irregularity and process legitimacy are related constructed diagnostics, so the figure should be read as an emergency-procedure profile rather than three independent validation targets.

Table 9: Selected simulation diagnostics for the emergency-review argument

Scenario	Rights	Emerg. irregularity	Emerg. downstream	Process legit.	Lower ct.	Cost	Down. move	Cost move	Main tradeoff
Stylized current U.S.-like court	0.634	0.308	0.187	0.345	0.513	0.152	none	none	Higher claimant success but higher emergency irregularity
Mandatory written emergency reasoning	0.618	0.077	0.129	0.490	0.535	0.229	large better	large worse	Transparency gains with modest merits-work shift
Automatic merits follow-up	0.632	0.030	0.089	0.547	0.541	0.245	large better	large worse	Regularizes emergency relief through merits routing
No emergency relief without merits review	0.633	0.022	0.088	0.573	0.543	0.230	large better	large worse	Lowest emergency irregularity; watch claimant routes
Independent recusal enforcement with substitutes	0.622	0.107	0.114	0.520	0.535	0.235	large better	large worse	Reduces recusal pressure with substitution costs
Judicial electorate selection court	0.624	0.108	0.114	0.527	0.535	0.277	large better	large worse	Appointment insulation, not emergency reform
Randomized merits panels with en banc correction	0.622	0.107	0.114	0.545	0.536	0.237	large better	large worse	Access/correction tradeoff through en banc review
Emergency integrity package	0.634	0.030	0.090	0.577	0.543	0.268	large better	large worse	Broad emergency-process package with higher complexity
Public-interest litigation filter	0.628	0.106	0.113	0.511	0.536	0.238	large better	large worse	Filters access while preserving public-law claims

Movement labels compare each design with the current-like baseline; no material movement means $|\Delta| < .010$, small means $.010 \leq |\Delta| < .050$, and large means $|\Delta| \geq .050$.

carrying higher emergency irregularity. Written reasons, automatic merits follow-up, and no-relief-without-merits review reduce emergency irregularity or downstream effects more directly than judicial-electorate selection or randomized panels, which are better understood as appointment-insulation or routing/correction mechanisms. The cost column is included in the main table because emergency regularity is often purchased with added procedure, merits acceleration, or docket-processing load.

9.1 The Price of Emergency Regularity

Emergency regularity is not free in the reported simulations. Reason-giving requirements add explanation costs, automatic merits follow-up moves emergency pressure into the merits docket, and no-relief-without-merits rules can preserve procedural discipline only if urgent claimants still have a timely merits route. Those costs are visible in administrative cost, merits acceleration, emergency-rights claimant success, and lower-court compliance rather than in emergency irregularity alone. The institutional lesson is therefore not that emergency restraint is uniformly superior; it is that rules that discipline emergency relief should be evaluated by their downstream disruption and capacity demands.

Table 10: Mechanism-level paired contrasts, interpreted by movement thresholds

Mechanism	Emerg. irregularity	Emerg. downstream	Rights	Lower ct.	Cost	Reading
Emergency restraint	large improvement (-0.314)	large improvement (-0.113)	no material movement (-0.001)	small improvement (+0.031)	large deterioration (+0.083)	emergency-process mechanism
Mandatory written emergency reasoning	large improvement (-0.249)	large improvement (-0.064)	small deterioration (-0.022)	small improvement (+0.022)	large deterioration (+0.080)	emergency-process mechanism
Automatic merits follow-up	large improvement (-0.301)	large improvement (-0.111)	no material movement (-0.005)	small improvement (+0.031)	large deterioration (+0.096)	emergency-process mechanism
Strong recusal enforcement	large improvement (-0.218)	large improvement (-0.081)	small deterioration (-0.019)	small improvement (+0.023)	large deterioration (+0.086)	recusal and process legitimacy
Randomized merits panels	no material movement (+0.001)	no material movement (+0.000)	no material movement (+0.002)	no material movement (+0.000)	small deterioration (+0.046)	routing and correction mechanism
Constitutional remand	no material movement (-0.001)	no material movement (-0.001)	no material movement (-0.001)	small improvement (+0.019)	large deterioration (+0.094)	post-decision response mechanism
Public-interest litigation filter	no material movement (-0.002)	no material movement (-0.003)	no material movement (+0.006)	no material movement (+0.002)	small deterioration (+0.024)	mixed institutional mechanism
Judicial electorate selection	no material movement (+0.001)	no material movement (+0.000)	no material movement (-0.001)	no material movement (-0.000)	large deterioration (+0.063)	appointment insulation comparison
Legislative override window	no material movement (+0.000)	no material movement (+0.001)	no material movement (+0.003)	no material movement (-0.003)	small deterioration (+0.030)	post-decision response mechanism
Jurisdiction-stripping constraints	no material movement (+0.001)	no material movement (+0.001)	no material movement (-0.001)	no material movement (+0.000)	small deterioration (+0.021)	mixed institutional mechanism
Emergency integrity bundle	large improvement (-0.302)	large improvement (-0.112)	no material movement (-0.000)	small improvement (+0.030)	large deterioration (+0.121)	emergency-process mechanism
Remand and override-window bundle	small improvement (-0.030)	small deterioration (+0.020)	no material movement (-0.006)	small improvement (+0.017)	large deterioration (+0.149)	post-decision response mechanism
Panel and jurisdiction safeguards	no material movement (-0.007)	small deterioration (+0.022)	no material movement (-0.009)	no material movement (-0.002)	large deterioration (+0.070)	routing and correction mechanism
Council with concrete-review backstop	no material movement (-0.001)	no material movement (-0.001)	no material movement (+0.004)	no material movement (+0.006)	large deterioration (+0.100)	mixed institutional mechanism

Thresholds: no material movement is $|\Delta| < .010$; small movement is $.010 \leq |\Delta| < .050$; large movement is $|\Delta| \geq .050$.
Cost deterioration means higher modeled administrative cost.

Table 10 reports the mechanism-level paired contrasts with movement thresholds instead of a long decimal grid. Emergency restraint, written emergency reasoning, automatic merits follow-up, and the emergency-integrity bundle are the rows most directly connected to the paper’s claim. Strong recusal enforcement, judicial-electorate selection, randomized panels, constitutional remand, public-interest filtering, jurisdiction-stripping constraints, and legislative override windows remain important because they test whether non-emergency mechanisms reduce arbitrary emergency power or merely move costs into access, compliance, administrative burden, or conflict.

10 Sensitivity and Robustness

Table 11 reports named-prior uncertainty bands from the parameter sweep. The overlap among the leading designs is the main point. The model is more informative about tradeoff patterns, such as emergency-restraint reducing emergency irregularity and dual-court designs increasing conflict under the current disagreement rule, than about fine ordinal rankings among close scores.

Table 11: Simulation scores and uncertainty bands for selected designs

Scenario	Avg. score	Score 5/50/95	Irregularity 5/50/95	Conflict 5/50/95
18-year terms	0.565	0.514/0.606/0.621	0.059/0.078/0.159	0.339/0.382/0.588
Written emergency reasons	0.561	0.508/0.602/0.618	0.033/0.049/0.123	0.339/0.385/0.594
Emergency merits follow-up	0.560	0.503/0.604/0.619	0.012/0.017/0.053	0.340/0.376/0.596
No emergency merits gap	0.565	0.512/0.609/0.625	0.007/0.011/0.043	0.342/0.376/0.594
Current-like	0.552	0.498/0.590/0.609	0.177/0.238/0.423	0.361/0.410/0.644
Strong recusal enforcement	0.559	0.509/0.598/0.615	0.057/0.078/0.163	0.344/0.384/0.591
Judicial electorate selection	0.554	0.500/0.595/0.612	0.057/0.078/0.162	0.342/0.386/0.596
Constitutional remand	0.566	0.514/0.601/0.617	0.057/0.078/0.166	0.341/0.380/0.593

Table 12: Sampled-prior uncertainty bands for selected designs

Scenario	Score	Rights	Emerg. irregularity	Emerg. down.	Lower ct.	Gov. noncomp.	Conflict	Interpretation
No emergency relief without merits review	0.596/0.615/0.630	0.638/0.649/0.657	0.005/0.010/0.016	0.033/0.058/0.110	0.585/0.605/0.623	0.043/0.066/0.101	0.315/0.358/0.414	leading group
18-year staggered terms + regular appointments	0.589/0.611/0.628	0.630/0.642/0.652	0.052/0.073/0.109	0.043/0.071/0.128	0.572/0.597/0.615	0.040/0.068/0.103	0.318/0.361/0.420	leading group
Automatic merits follow-up for emergency relief	0.592/0.610/0.625	0.635/0.649/0.658	0.009/0.015/0.025	0.043/0.067/0.126	0.582/0.604/0.623	0.045/0.063/0.108	0.318/0.368/0.422	leading group
Mandatory written emergency reasoning	0.586/0.607/0.623	0.625/0.641/0.650	0.029/0.046/0.074	0.049/0.076/0.141	0.574/0.596/0.617	0.048/0.072/0.107	0.321/0.370/0.423	leading group
Constitutional remand before invalidation	0.585/0.607/0.623	0.626/0.640/0.650	0.053/0.076/0.111	0.044/0.072/0.129	0.587/0.610/0.629	0.044/0.067/0.106	0.318/0.365/0.419	leading group
Independent recusal enforcement with substitutes	0.584/0.606/0.624	0.632/0.642/0.652	0.053/0.075/0.112	0.046/0.073/0.134	0.574/0.597/0.617	0.045/0.072/0.106	0.320/0.367/0.424	leading group
Randomized merits panels with en banc correction	0.583/0.605/0.622	0.632/0.641/0.651	0.053/0.074/0.109	0.045/0.071/0.126	0.574/0.597/0.615	0.043/0.069/0.109	0.320/0.366/0.414	leading group
Judicial electorate selection court	0.579/0.600/0.617	0.629/0.645/0.651	0.052/0.075/0.111	0.045/0.073/0.130	0.574/0.598/0.615	0.046/0.071/0.112	0.319/0.371/0.424	leading group
Stylized current U.S.-like supreme court	0.568/0.594/0.616	0.628/0.645/0.660	0.168/0.232/0.324	0.080/0.123/0.197	0.541/0.577/0.603	0.049/0.098/0.153	0.338/0.399/0.466	overlapping uncertainty band

Table 12 adds a stricter uncertainty check: instead of only changing named assumption cases, it samples prior distributions over polarization, appointment capture, public pressure, emergency share, justice-pool size, and legislative-output profile components. Overlapping uncertainty bands are interpreted as mechanism contrasts, not rankings.

Table 13: Sensitivity drivers and interpretation caveats

Prior	Top-cluster design	Score	Rights	Emerg. irregularity	Emerg. down.	Gov. noncomp.	LC resist.	Caveat
Baseline institutional prior	No emergency relief without merits review	0.610	0.646	0.011	0.055	0.072	0.245	leading group
Baseline institutional prior	60 percent invalidation threshold	0.609	0.634	0.067	0.080	0.067	0.248	rights-protection caveat
High emergency-share prior	18-year staggered terms + regular appointments	0.520	0.618	0.177	0.219	0.223	0.402	compliance caveat
High emergency-share prior	Jurisdiction stripping constrained by rights carveouts	0.517	0.621	0.185	0.223	0.247	0.404	compliance caveat
High rights-risk prior	No emergency relief without merits review	0.526	0.602	0.027	0.098	0.228	0.398	compliance caveat
High rights-risk prior	Constitutional remand before invalidation	0.526	0.565	0.118	0.131	0.215	0.363	compliance caveat
Weak democratic-mandate prior	60 percent invalidation threshold	0.544	0.604	0.095	0.127	0.184	0.378	compliance caveat
Weak democratic-mandate prior	Constitutional remand before invalidation	0.540	0.616	0.108	0.111	0.173	0.338	compliance caveat
High constitutional-conflict prior	60 percent invalidation threshold	0.513	0.602	0.143	0.194	0.233	0.438	compliance caveat
High constitutional-conflict prior	Constitutional remand before invalidation	0.509	0.616	0.155	0.168	0.237	0.400	compliance caveat
Imported legislative-family prior	No emergency relief without merits review	0.618	0.655	0.008	0.054	0.057	0.229	leading group
Imported legislative-family prior	18-year staggered terms + regular appointments	0.615	0.643	0.069	0.065	0.064	0.233	rights-protection caveat

Table 13 identifies the assumption cases that drive the top cluster. High-emergency and high-conflict priors make emergency and conflict mechanisms more influential, while rights, compliance,

lower-court-resistance, and emergency-power caveats identify the tradeoffs that carry through the sweep.

11 Limitations

Five limitations matter most. First, the parameter weights are not estimated coefficients. They are literature-informed and design-prior coding choices tested through shared-world comparisons and sensitivity sweeps. Second, the main emergency-irregularity result is partly endogenous to the model design: written reasons, merits follow-through, and merits acceleration are components of the emergency-irregularity formula, so reductions in that index should be read as design implications rather than independent findings. The more probative results are downstream changes in emergency disruption, rights-claimant success, lower-court compliance, government noncompliance, conflict, and administrative cost. Third, denominator matches remain weak for several pathways, especially emergency applications, lower-court compliance, comparative access paths, and override/remand behavior. The paired OT2023-OT2024 certiorari extracts close count-defined paid/IFP docketed-intake denominators and direct docket-visible CFR and CVSG fields for two terms, but they exclude undocketed submissions, are not constitutional-review-only, leave issue/counsel/split/vehicle coding open, and retain partial petitioner/respondent and merits fields. Three OT2024 certiorari outcomes remain pending or held, and the OT2024 snapshot excludes four same-cutoff-date dockets above the Journal count-defined ranges. The precedent-treatment extract improves aggregate doctrinal-uptake evidence. The environmental snapshot adds 191 published citation-linked opinions, 65 nationwide-applicability/citation-presence cells, and five practical episodes, but it is not an all-relevant-case opportunity or ignored-precedent denominator. Only 115 opinions have public-text context; this missingness is nonrandom, the six directional values are automated candidates pending expert review, the decisions are purposively selected statutory cases, remedy fidelity remains uncoded, and the practical sample contains no noncompliant outcome. A HUDOC-EXEC extract separately supplies bounded ECtHR execution-monitoring evidence for the monitoring-capacity pathway, but it does not validate lower-court compliance, implementation resistance, government noncompliance, or emergency downstream effects. Fourth, process legitimacy and constitutional conflict are constructed indices that combine several legal and political channels. Fifth, observed reform benchmarks are still needed before the model can be used for stronger empirical claims.

The strongest next validation steps are to complete independent expert coding and adjudication for the environmental review queue; automate raw-data refresh for SCDB, Harvard Law Review statistics, shadow-docket data, and recusal data; extend the certiorari cohort beyond two terms and code respondent type, issue area, counsel specialization, alleged and genuine splits, vehicle quality, SG response and recommendation, and merits follow-through; calibrate amparo, constitutional complaint, QPC, abstract-review, and concrete-referral denominators from raw sources; expand post-decision coding beyond the five environmental cases to a relevant-case opportunity and remedy-fidelity denominator with observed resistance and noncompliance outcomes; add observed counterfactual benchmarks where institutional changes can be compared before and after reform; and benchmark the imported legislative-output profile against raw legislative simulator outputs rather than only aggregate CSV summaries.

Table 14 converts the denominator audit into a benchmark agenda. It is a work queue, not evidence of validation: rows remain limitations until the listed source extraction or benchmark comparison is actually added.

The associated extraction protocol in the replication materials specifies the emergency-application fields needed to evaluate merits follow-through, reason visibility, and downstream implementation

Table 14: Benchmark-readiness agenda for stronger validation

Rank	Pathway	Construct	Current gap	Required evidence	Manuscript use
1	certiorari	elite counsel access	needs direct behavior benchmark	same-row petition representation coding for specialist or former-clerk counsel, represented side, disposition, and denominator rule	not validation yet
2	certiorari	genuine split quality	needs pathway-specific denominator	same-row alleged and genuine split, split depth, issue, vehicle quality, disposition, and denominator rule	not validation yet
3	complaint referral	filtered referral merits path	needs count-to-rate denominator	pathway-specific filing, admission, decision, and remedy time series from official constitutional-court statistics	not validation yet
4	emergency	emergency application presence	needs pathway-specific denominator	application-level emergency dataset linking applicant, relief request, reason-giving, grant, merits follow-through, and downstream status	not validation yet
6	lower court compliance	implementation resistance	needs broader outcome and noncompliance coverage	broader decision and agency coverage with a relevant-case opportunity denominator, remedy fidelity, and observed resistance or noncompliance outcomes while preserving the existing event, practical-implementation, and monitoring source boundaries	use the completed bounded source slice only; do not generalize to denominator-matched validation
7	lower court compliance	monitoring capacity	bounded monitoring slice; needs broader coverage	broader decision and agency coverage with a relevant-case opportunity denominator, remedy fidelity, and observed resistance or noncompliance outcomes while preserving the existing event, practical-implementation, and monitoring source boundaries	use the completed bounded source slice only; do not generalize to denominator-matched validation
8	complaint referral	constitutional remand/deferred remedy	needs direct behavior benchmark	pathway-specific filing, admission, decision, and remedy time series from official constitutional-court statistics	not validation yet
15	override remand	rights carveout pressure	needs direct behavior benchmark	event catalog distinguishing invalidation, delayed remedy, legislative response, override attempt, and override success	not validation yet

The agenda is generated from the denominator audit and is not itself validation evidence; it identifies source extractions required before stronger empirical language would be justified.

effects on an application-level denominator. Official-docket coded benchmarks cover all 31 granted source-record rows, representing 30 unique emergency matters across OT2023 and OT2024, and all 210 denied or non-binary source-record rows. The denied/non-binary slice adds application filing dates for all 210 rows, 60 court-requested response rows, 40 docket-visible reason-marker rows, and 55 repeat-filing rows, with no possible merits-follow-through rows remaining under a review-required code. One mixed-disposition source row is manually linked to a later merits docket that was granted and remained pending at the July 26, 2026 refresh. These materials support bounded all-application docket-linkage diagnostics; they do not support external implementation claims.

The human-review handoff contains separate independent-reader forms, source-linked case cards, an adjudication form, and a validator for returned coding. All 16 initial environmental review rows remain pending human judgment. A separate acquisition queue maps the 241 emergency source records to 227 term/docket matters while preserving multiple order events. The four historical Journal fetch failures are cross-referenced to observations already present in the later OT2023 cohort. Neither these administrative reconciliations nor the independent AI methods review supplies human signoff. The delivered replication archive is tested by extracting it, rebuilding the analytic workflow, and comparing deterministic report and manuscript inputs.

For certiorari, independently enumerated official-docket extracts close 4,222 OT2023 and 3,854 OT2024 paid/IFP dockets with zero failed fetches and identify 4,033 and 3,683 certiorari petitions, respectively. The OT2023 cohort has mature current outcomes; the OT2024 cohort retains three pending or held petitions and computes its grant/GVR rate over the 3,680 resolved rows. Both supply docket-visible filing, paid/IFP, response/CFR, CVSG, amicus-presence, relist, and cert-stage outcome fields, subject to structurally conditional blanks. The OT2024 manifest records four same-cutoff-date dockets above the Journal count-defined ranges as explicit snapshot exclusions. In the OT2023 independent reconciliation, current docket outcomes match the parsed Journal

disposition slice for 3,290 dockets; 932 cohort dockets are absent from that older disposition-defined extract, and no outcome differences remain. The older 3,951-row Journal disposition seed, 3,947-row Journal public-docket join, four-row failed-page workqueue, and 115-row granted/GVR join remain independent OT2023 reconciliation slices rather than cohort denominators. The field-readiness matrix keeps petitioner/respondent and several merits fields partial and leaves issue area, specialist/former-clerk counsel, alleged and genuine split quality, split depth, and vehicle quality uncoded. For implementation and compliance, the replication materials now include the 876-precedent aggregate treatment extract and its 223-precedent constitutional-issue subset; 191 published environmental citation events and 65 nationwide-applicability/citation-presence cells; a missingness audit and pending expert-review queue; five practical environmental case-study episodes; and a HUDOC-EXEC pending leading-case monitoring extract with 1,450 rows as of July 2, 2026. These support bounded aggregate uptake, descriptive citation-presence, purposive practical implementation, and monitoring-capacity language. They do not establish lower-court treatment validity, a relevant-case opportunity or ignored-precedent rate, representative constitutional or agency compliance, remedy fidelity, government noncompliance, or emergency downstream implementation.

A Institutional Parameter Provenance

Appendix Table 16 classifies major parameter families by the source of discipline behind the coding choice. The coefficients are not estimated; the table identifies which families have direct source support, which are design priors, and which require sensitivity or external benchmark work before stronger interpretation.

Appendix Table 17 records denominators and interpretation notes for representative metrics.

The anonymous technical supplement retains alternative priority scores (Table S1), full litigation and enforcement diagnostics (Table S2), and coding-versus-downstream interpretation rules (Table S3). These remain descriptive model outputs and assumptions, not empirical validation.

Figure 4 is an appendix diagnostic map. It plots the process-legitimacy index against the constitutional-conflict index for selected designs, with marker shade representing emergency irregularity. It should not be read as a two-dimensional welfare frontier because rights protection, domain-specific claimant outcomes, democratic responsiveness, and administrative cost are intentionally outside the display.

Table 15: Illustrative provenance map for modeled institutional features

Feature family	Model representation	Provenance status
Appointment and tenure	Appointment method, judicial-electorate selector pool, judicial-electorate nominee pool, confirmation threshold, vacancy deadlock risk, term renewability, retirement age, replacement timing	Comparative institutional coding and scenario assumptions
Review access	Access path, petition type, court-requested response, CVSG request, admission score, claimant type, bar capital, claim strength, vehicle quality, lower-court split depth, strategic plaintiff selection, repeat-player advantage, merits transfer, screen-out	Normalized source-register rows and modeled admission rules
Emergency procedure	Application class, relief request, response request, full-court referral, reason giving, status-quo effect, merits follow-through, downstream effect	Shadow-docket source tables plus scenario rules
Decision rule	Voting threshold, panel/en banc routing, quorum failure, recusal consequence, per-remedy thresholds	Court-design coding and institutional assumptions
Post-decision politics	Override attempt, rights carveout, formal response, practical implementation, government noncompliance, open defiance	Comparative public-law coding and strategic actor assumptions
Outputs	Stability, rights protection, claimant success, legitimacy, conflict, responsiveness, precedent durability, lower-court compliance	Simulated outcomes with calibration guardrails

Table 16: Parameter-justification framework

Parameter family	Source of discipline	Est.	Sensitivity tested?	Main risk
Admission weights	Certiorari and intake source-register patterns plus model priors	No	partial	overfits cert-like processes if comparative access paths are under-calibrated
Emergency irregularity	Emergency-procedure theory, shadow-docket scholarship, design prior	No	ablation table	headline movement can be partly mechanical
Process legitimacy	Procedural-legitimacy theory and process-visibility coding	No	penalty ablations	can inherit emergency-irregularity and partisan-alignment penalties
Rights and claimant success	Legal-domain profiles and rights-burden coding	No	domain heatmap	claimant success is not identical to rights protection
Compliance and conflict	Implementation literature, lower-court behavior, proxy rows	No	partial	denominator matches remain weak
Administrative cost	Institutional-complexity and docket-processing design prior	No	mechanism contrasts	can understate real institutional capacity constraints
Override/remand behavior	Weak-form and comparative constitutional-design literature	No	partial	cross-jurisdiction comparisons are not direct validation targets

Table 17: Metric semantics and manuscript use

Family	Metric	Use	Compatibility	Manuscript interpretation
certiorari	paid cert petition share	loose calibration	near pathway match	conditional paid/IFP split is closer than all-case paidPetitionRate, but the simulator still excludes many non-review docketed matters
certiorari	certiorari admission rate	proxy context	denominator mismatch	the source includes all docketed matters; the simulator denominator is already a constitutional-review cert-path subset
emergency	emergency grant conditional rate	loose calibration	conditional near match	conditional rate is closer than all-case emergencyGrantRate, but the simulated emergency universe is not limited to noncapital applications
complaint referral	admission rate	proxy context	denominator mismatch	this is a strong complaint-filter source but not a whole-universe admission target
lower court compliance	lower court compliance	proxy context	scale mismatch	direct observed aggregate doctrinal treatment is citation/treatment-selected and citation-weighted, lacks an exposed or ignored-case denominator, and is not the simulator's synthetic case-average score
lower court compliance	government noncompliance rate	loose calibration	mechanism proxy	closest available implementation anchor, but still a study-sample proxy rather than a court-wide noncompliance rate
override remand	override rate	design prior	design prior	duration informs institutional design, not observed behavioral override success
headline	directional score	reading aid	not empirical target	used only for ordering and clustering; close score differences are not substantive rankings
headline rights	rights claimant success	synthetic output	not empirical target	case-average claimant-success output
	emergency rights claimant success	synthetic output	not empirical target	conditional emergency-route claimant-success output
headline	emergency process irregularity	synthetic output	not empirical target	constructed emergency irregularity index
headline	process legitimacy proxy	synthetic output	not empirical target	constructed process-legitimacy index
headline	constitutional conflict	synthetic output	not empirical target	mechanism summary metric; individual channels remain reported separately

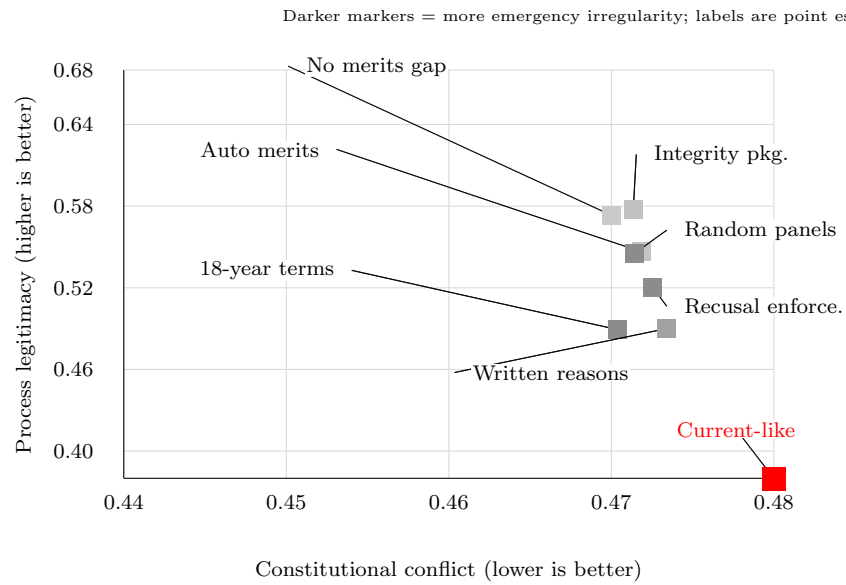


Figure 4: Appendix diagnostic map of process legitimacy and constitutional conflict for selected institutional designs. Marker shade increases with emergency irregularity.

B Model Mechanics

This appendix records the main state machine and scoring formulas so the manuscript does not treat the simulator as a black box. Each run draws a filed universe from a shared world seed. For each scenario, the same filed matters are reviewed by a court process with a scenario-specific design and a scenario-specific random stream. Let i index filed matters and s index institutional scenarios.

Admission begins with a path-specific petition probability. In the compact notation below, c is legal conflict, a is public attention, g is SG signal, d is split depth, k is bar capital, m is claim strength, r is repeat-player advantage, p is strategic-plaintiff selection, f is forum-shopping pressure, q is settlement pressure, and v is vehicle-defect risk:

$$P_{is} = \text{clamp}_{[0,1]}(b_{\text{petition}} + .08c_i + .08a_i + .06g_i + .08d_i \\ + .05k_i + .05m_i + .06r_i + .08p_i + .04f_i - .05q_i - .22v_i).$$

Conditional on a petition or referral, the admission score combines certiorari pressure, lower-court conflict, lower-court error, lower-court split depth, genuine split status, SG signal, court-requested response, CVSG request, amicus intensity, split maturity, relist signal, claimant type, bar capital, claim strength, vehicle quality, specialist counsel, repeat-player advantage, strategic plaintiff selection, forum-shopping pressure, settlement pressure, conditional reversal probability, access-path adjustment, institutional gatekeeping, vehicle defects, and vacancy deadlock. Merits transfer is then path-specific: emergency applications transfer through the application’s merits-follow-through probability, complaints and amparo transfer at a lower rate, and ordinary referrals or cert-like paths transfer at a higher baseline reduced by settlement pressure.

Table S4 in the anonymous technical supplement records representative weights and their interpretation.

After admission, the court process chooses strategic-actor response, emergency status, pre-review settlement, initial merits review, panel or en banc routing, recusals, quorum, votes, emergency order, invalidation, auxiliary-review outcome, override attempt, and post-decision implementation response. Case-level metrics are averaged over the filed universe. For example, rights-claimant success is one when a high-rights-burden claimant receives invalidation or emergency relief and partial when settlement or constitutional remand preserves a claimant’s route without immediate invalidation; process legitimacy is legitimacy minus penalties for emergency legitimacy risk, partisan alignment, appointment-pressure campaigns, and court-curbing response; the constitutional-conflict index rises with conflict potential, executive defiance, lower-court conflict, lower-court resistance, accumulated state conflict, government noncompliance, emergency downstream effect, invalidating high-mandate laws, emergency relief, cross-court disagreement, override attempts, and repeated override loops.

The directional score is a secondary, normative reading aid. The technical supplement records its formula and component weights alongside Table S1. Differences below .010 are treated as indistinguishable; the substantive analysis emphasizes individual outcomes and tradeoffs.

Data Availability Statement

The replication package includes simulator code, build instructions, normalized input data, deterministic seeds, scenario definitions, generated reports, manuscript figures, the source-audit file (`paper/source-audit.csv`), provenance manifests, and artifact hashes. Public repository identifiers are withheld in this anonymous review manuscript. For final publication, the package should be deposited in the Journal of Law and Courts Dataverse or another repository linked from the JLC Dataverse.

Funding Statement

No external funding is reported for this version of the manuscript.

Competing Interests

Competing interests: The author(s) declare none.

AI Assistance Statement

The project used OpenAI ChatGPT Deep Research and OpenAI Codex from May through September 2026 to assist with research organization, source-synthesis review, data extraction and quality checks, code generation, drafting, formatting, local build verification, and preparation of human-review materials. These tools are accessible through OpenAI’s ChatGPT and Codex interfaces. The replication package records tools and use periods, not a complete session-by-session model-version history. An independent AI methods review is disclosed separately from the still-pending human legal coding and methods assessment. The author(s) remain responsible for reviewing all AI-assisted outputs, verifying sources and project artifacts, and approving the manuscript, simulator, calibration choices, figures, references, and claims. No AI-generated visual content appears in the manuscript; figures are generated deterministically from project CSV outputs.

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